

Riemann Sums and Definite Integrals

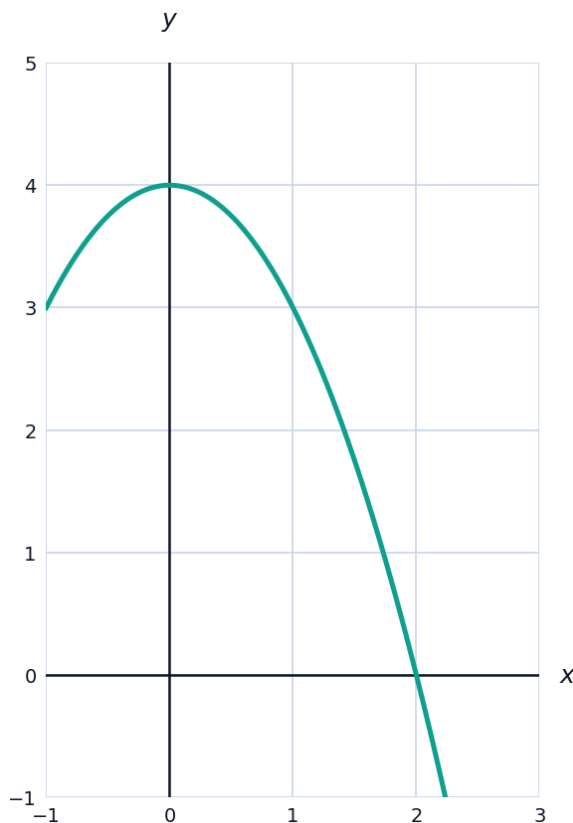


Approximating area with Riemann sums

- 1 Estimate $\int_0^4 x^2 dx$ using a right-endpoint Riemann sum with $n = 4$ subintervals.
- 2 Estimate $\int_1^5 (2x + 1) dx$ using a left-endpoint Riemann sum with $n = 4$ subintervals.

The definite integral and its properties

- 3 Given $\int_0^3 f(x) dx = 8$ and $\int_3^7 f(x) dx = -2$, find:
a $\int_0^7 f(x) dx$ **b** $\int_7^3 f(x) dx$ **c** $\int_0^3 5f(x) dx$
- 4 The graph of $f(x) = 4 - x^2$ is shown, together with the region between the curve and the x -axis on $[0, 2]$. Evaluate $\int_0^2 (4 - x^2) dx$ exactly.



- 5 Evaluate $\int_{-2}^2 (x^3 - 4x) dx$ by recognizing the integrand's symmetry, without computing an antiderivative.